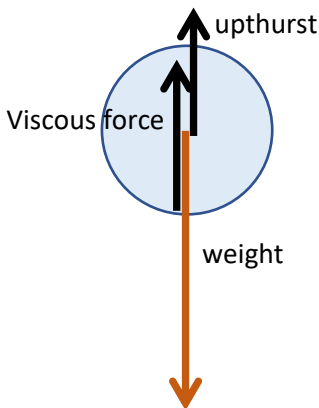
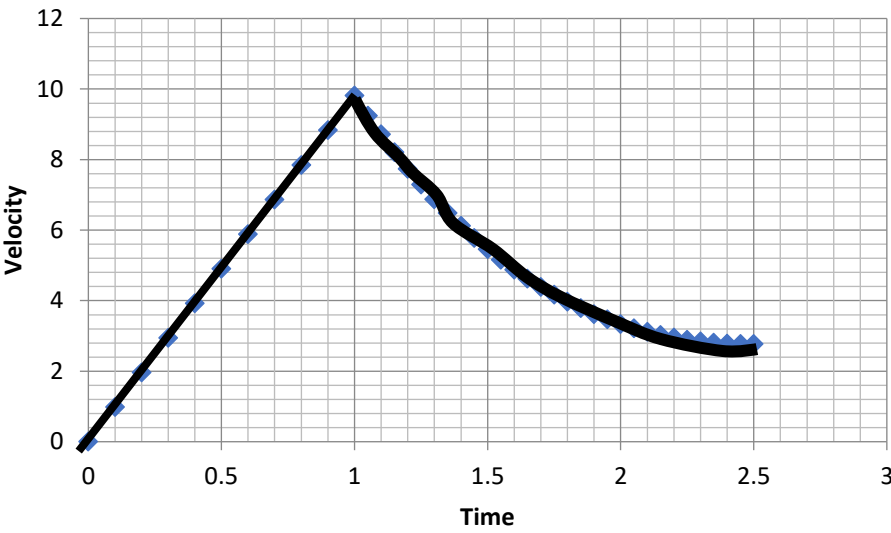


6ai	It is rate of change of velocity.	A1
6aii	The highest point of the motion when the ball is thrown vertically upwards/ object in simple harmonic motion at amplitude position OR any other reasonable scenario.	B1
6bi	Air resistance is negligible as the <u>acceleration of the ball when it is falling through air during 0s to 1s is constant at 9.8 ms^{-2}</u> , which shows that the <u>net force acting on the ball is weight of ball only</u> . Or Since acceleration is 9.8 m s^{-2} which is equal to the gravity acceleration, the net force acting on ball is weight only.)	A1
6bii	Zero acceleration at 2.5s does not mean that it had reach the bottom of the beaker, the <u>ball could have reached terminal velocity</u> where the drag force and upthrust balances the weight of the ball. Or <u>At 2.5 s, the velocity of the ball is not zero</u> (Area under acceleration-time graph gives us the change of velocity. Since area under the graph of acceleration time graph from $t=0$ to $t=2.5\text{s}$ is not zero, velocity at 2.5 s is not zero). Hence, <u>if the ball had reached the bottom of the beaker, there should be a large negative acceleration upon impact</u> .	A1
6biii	Upward force: viscous force and upthrust, downward force: weight. Length of the forces should be balanced. 	A1
6biv	Highest velocity = area under acceleration-time graph for t between 0 to 1s, = $(9.81)(1) = 9.81 \text{ ms}^{-1}$. Or using $v = u + at$ where $u=0$ $v = 0 + 9.81(1)$ M1 = 9.81 ms^{-1} A1	M1 A1
6bv	Terminal velocity occurs at $t=2.5\text{s}$. Terminal velocity = area under acceleration-time graph for t between 0 to 3s, = $9.81 - \text{area under acceleration-time graph for t between 1 to 3s}$ = $9.81 - 73(1)(0.1)$ = 2.51 ms^{-1}	M1 A1

6bvi	 Correct shape and value for each of the segment For 0 s to 1 s For 1 s to 2.5 s	B1 B1
6ci	<u>Speed of trolley after collision, v_f can be obtained from conservation of energy.</u> [Gain in gravitational of potential energy of trolley with block and pellet equal to loss in Kinetic Energy of trolley with block and pellet after collision. i.e. $Mgh = \frac{1}{2} Mv_f^2$, $v_f = \sqrt{2gh}$] <u>By conservation of linear momentum, initial momentum of pellet = final momentum of trolley with block and pellet, the speed of the pellet can be determined.</u> [i.e. $mu = Mv_f \rightarrow$ Speed of pellet, $u = \frac{M}{m}v_f$]	B1 B1
6cii	u is calculated assuming all kinetic energy of trolley with block and pellet after collision is converted to gravitational potential energy. With presence of friction, <u>some of the kinetic energy of trolley after collision is converted to work done against frictional force.</u> Since h is a fixed measurement, <u>u calculated is an underestimate.</u>	M1 A1
6di	When objects of a closed system interact, their sum of momentum before impact is equal to their sum of momentum after impact, if no net external force acts on the system. [Only need to mention either closed system or no net external force. Need not mention both.]	A1
6dii	By conservation of linear momentum, $P_i = P_f$ $0.0020(u) = (0.5020)(0.4)$ $u = 100.4 \text{ ms}^{-1} = 100 \text{ ms}^{-1}$	M1 A1
6e	Kinetic Energy of pellet before hitting trolley with block $= \frac{1}{2} (0.0020)(100^2) = 10.0 \text{ J}$ Kinetic Energy of trolley with block and pellet after collision $= \frac{1}{2} (0.5020)(0.4^2) = 0.0402 \text{ J}$	M1 (for 2 KE)

	Since Kinetic Energy before and after collision are not the same value, the collision between pellet and block is not inelastic. OR By relative speed of relation, speed of approach is non-zero while speed of separation is zero (as the bullet is embedded)	A1
7a	Temperature of a substance is directly proportional to the average kinetic energy of the	B1